Lecture plan · for the lecturer

Privacy attacks and differential privacy

AI security · University · Lecture 3

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Explain membership inference.
- Define differential privacy.
- Calibrate noise.

Activity: Respond at random

1. Warm-up (5 min): Show the de-identified ratings and ask how someone could be recognised.
2. Flip (10 min): Students answer a sensitive question by randomised response.
3. Estimate (15 min): Recover the true rate from the noisy answers.
4. Calibrate (25 min): Compute Laplace scales for several ε and compare released counts.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Why can no one be blamed for an answer?
- What does ε cost?
- Who should set the budget?

Exit ticket

- Which data release in your institution would you protect with differential privacy?

Try it in your next lecture
Ask the class a sensitive yes-or-no question by randomised response, with a coin flip first; estimate the true rate together and discuss why no single answer reveals anyone.

Seminar sheet · for students

Privacy attacks and differential privacy

AI security · University · Lecture 3

Name _____ Group _____ Date _____

1 **Which step gives a provable limit on membership leakage?**

◯ Removing names ◯ Differentially private training ◯ Early stopping

2 **Set the noise.**

A count has sensitivity 1. The Laplace scale is _____ for ε = 0.1 and _____ for ε = 1.

Word bank: 0.1 · 1 · 10 · 100

3 **Which statements about membership inference are true? Choose all that apply.**

☐ Removing names from training data prevents it

☐ Differential privacy bounds the attack's advantage

☐ A larger train-test gap usually helps the attacker ☐ Shadow models can imitate the target

4 **Answering two queries, each with ε = 0.5, gives a total privacy loss of at most ε = 1.**

◯ True ◯ False

5 **Order one DP-SGD step.**

☐ Clip each to norm C

☐ Update and account for ε

☐ Compute per-example gradients

☐ Add Gaussian noise to the sum

Answers · for the lecturer

Privacy attacks and differential privacy

AI security · University · Lecture 3

1 **Differentially private training**

It bounds every person's influence.

2 **A count has sensitivity 1. The Laplace scale is 10 for ε = 0.1 and 1 for ε = 1.**

Scale = Δ/ε: 1/0.1 = 10 and 1/1 = 1. Smaller ε means more noise and more privacy.

3 **A larger train-test gap usually helps the attacker; Shadow models can imitate the target; Differential privacy bounds the attack's advantage**

The model leaks through its behaviour, not through names.

4 **True**

Basic composition: privacy losses add.

5 **Top to bottom: 2, 4, 1, 3**

Clipping bounds sensitivity so the noise can be calibrated.

For you to think about

How do your students currently learn that removing names is not enough?